

Banker's Algorithm

Process	Allocation			Need			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	4	3	2	3	0
P1	3	0	2	0	2	0	2	4	0
P2	3	0	2	4	0	0	5	4	2
P3	3	1	1	0	1	1	4	4	4
P4	0	0	2	4	3	1	10	5	7
Total	10	5	7						

The request is 2, 3, 1 safe

2

a)

	Allocation	Max	(work) Available	Max-allocation Need
	A B C D	A B C D	A B C D	A B C D
P0	2 1 0 1	4 2 1 2	3 3 2 1	2 1 1 1
P1	3 1 2 1	5 2 5 2	5 4 2 2	2 1 3 1
P2	2 1 1 3	2 3 1 6	6 7 3 4	0 2 0 3
P3	1 3 1 2	1 4 2 4	7 1 6 6	0 1 1 2
P4	1 4 3 2	3 6 6 5	10 12 8 7	2 2 3 3
+	9 10 7 9		12 13 9 10	
+ 2 3 2 1				
	11 13 9 10			

Safe sequence: < P0, P3, P4, P2, P1 >

Banker's algorithm
 $Need \leq work \Rightarrow work = work + allocation$
 P0: $2 1 1 1 \leq 3 3 2 1 \checkmark$
 so $work = \langle 3 3 2 1 \rangle + \langle 2 1 0 1 \rangle$
 $work = \langle 5 4 2 2 \rangle$
 P1: $2 1 3 1 \leq 5 4 2 2 \times$
 P2: $0 2 0 3 \leq 5 4 2 2 \times$
 P3: $0 1 1 2 \leq 5 4 2 2 \checkmark$
 $work = \langle 5 4 2 2 \rangle + \langle 1 3 1 2 \rangle$
 $work = \langle 6 7 3 4 \rangle$
 P4: $\langle 2 2 3 3 \rangle \leq \langle 6 7 3 4 \rangle \checkmark$
 $work = \langle 6 7 3 4 \rangle + \langle 2 1 4 3 \rangle$
 $work = \langle 7 11 6 6 \rangle$
 P1: $\langle 2 1 3 1 \rangle \leq \langle 7 11 6 6 \rangle \checkmark$
 $work = \langle 7 11 6 6 \rangle + \langle 3 1 2 1 \rangle$
 $work = \langle 10 12 8 7 \rangle$
 P2: $\langle 0 2 0 3 \rangle \leq \langle 10 12 8 7 \rangle \checkmark$
 $work = \langle 10 12 8 7 \rangle + \langle 2 1 1 3 \rangle$
 $work = \langle 12 13 9 10 \rangle$

b) If request from process P1 arrives for (0, 0, 1, 0) can the request be granted immediately?

	Allocation	Max	Available	Max-allocation Need
	A B C D	A B C D	A B C D	A B C D
P0	2 1 0 1	4 2 1 2	3 3 2 1	2 1 1 1
P1	3 1 2 1	5 2 5 2		2 1 3 1

- Request $\leq$ need
 $0 0 1 0 \leq 2 1 3 1 \checkmark$
- Request $\leq$ available
 $0 0 1 0 \leq 3 3 2 1 \checkmark$
- Available = Available - Request
 $= 3 3 2 1 - 0 0 1 0$
 $= 3 3 1 1$
 Allocation = Allocation + Request
 $= 3 1 2 1 + 0 0 1 0$
 $= 3 1 3 1$
 Need = Need - Request
 $= 2 1 3 1 - 0 0 1 0$
 $= 2 1 2 1$

New table

	Allocation	Max	(work) Available	Need
	A B C D	A B C D	A B C D	A B C D
P0	2 1 0 1	4 2 1 2	3 3 1 1	2 1 1 1
P1	<u>3 1 3 1</u>	5 2 5 2	5 4 1 2	<u>2 1 2 1</u>
P2	2 1 1 3	2 3 1 6	6 7 2 4	0 2 0 3
P3	1 3 1 2	1 4 2 4	4 8 5 5	0 1 1 2
P4	1 4 3 2	3 6 6 5	7 11 6 8	2 2 3 3

Safe sequence: < P0, P3, P4, P2, P1 >

9 10, 6, 9
 Total 12 13 9 10
 Banker's algorithm

- P0: Need $\leq$ work
 $2 1 2 1 \leq 3 3 1 1 \checkmark$
 $work = 3 3 1 1 + 2 1 0 1$
 $work = 5 4 1 2$
 P1: Need $\leq$ work
 $2 1 2 1 \leq 5 4 1 2 \times$
 P2: $0 2 0 3 \leq 5 4 1 2 \times$
 P3: $0 1 1 2 \leq 5 4 1 2 \checkmark$
 $work = 5 4 1 2 + 1 3 1 2$
 $work = 6 7 2 4$
 P4: $2 2 3 3 \leq 6 7 2 4 \times$
 P1: $2 1 2 1 \leq 6 7 2 4 \checkmark$
 $work = 6 7 2 4 + 3 1 3 1$
 $work = 9 8 5 5$
 P2: $0 2 0 3 \leq 9 8 5 5 \checkmark$
 $work = 9 8 5 5 + 2 1 1 3$
 $work = 11 9 6 8$
 P3: $2 2 3 3 \leq 11 9 6 8 \checkmark$
 $work = 11 9 6 8 + 1 4 3 2$
 $work = 12 13 9 10$

Yes, it can be served

Q3

Consider the following snapshot of a system:

Process	Allocation	Max	Need	Available
	A B C D	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	0 0 0 0	7 5 2 2
P1	1 0 0 0	1 7 5 0	0 7 5 0	1 5 3 2
P2	1 3 5 9	2 3 5 6	1 0 0 2	2 8 8 6
P3	0 6 3 2	0 6 5 2	0 0 2 0	2 14 11 8
P4	0 0 1 4	0 6 5 6	0 6 4 2	<u>3 14 12 12</u>

Answer the following:
 A) What is the content of need matrix?
 B) Is the system in a safe state?
 C) If a request from process P1 arrives for (0, 4, 2, 0) can the request be granted immediately?

Safe sequence: < P0, P1, P3, P4, P2 >

First, use banker's algorithm on original snapshot to determine if it is safe.

Need $\leq$ work $\Rightarrow$ work = work + allocation

- P0: $0 0 0 0 \leq 1 5 2 0 \checkmark$
 $work = 1 5 2 0 + 0 0 1 2$
 $work = 1 5 3 2$
 P1: $0 7 5 0 \leq 1 5 3 2 \times$
 P2: $1 0 0 2 \leq 1 5 3 2 \checkmark$
 $work = 1 5 3 2 + 1 3 5 9$
 $work = 2 8 8 6$
 P3: $0 0 2 0 \leq 2 8 8 6 \checkmark$
 $work = 2 8 8 6 + 0 6 3 2$
 $work = 2 14 11 8$
 P4: $0 6 4 2 \leq 2 14 11 8 \checkmark$
 $work = 2 14 11 8 + 0 0 1 4$
 $work = 2 14 12 12$
 P1: $0 7 5 0 \leq 2 14 12 12 \checkmark$
 $work = 2 14 12 12 + 1 0 0 0$
 $work = 3 14 12 12$

Yes, it is in a safe state.

c) Assume request fulfilled: P1 < 0, 4, 2, 0 >

Process	Allocation	Max	Need	Available
	A B C D	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	0 0 0 0	1 5 3 2
P1	1 4 2 0	1 7 5 0	0 3 3 0	1 1 2 2
P2	1 3 5 9	2 3 5 6	1 0 0 2	2 8 8 6
P3	0 6 3 2	0 6 5 2	0 0 2 0	2 14 11 8
P4	0 0 1 4	0 6 5 6	0 6 4 2	<u>3 14 12 12</u>

Should total: 3 14 12 12

Safe sequence: < P0, P2, P3, P4, P1 >

begin banker's need $\leq$ work $\Rightarrow$ work + allocation

- P0: $0 0 0 0 \leq 1 1 0 0 \checkmark$
 $work = 1 1 0 0 + 0 0 1 2$
 $work = 1 1 1 2$
 P1: $0 3 3 0 \leq 1 1 1 2 \times$
 P2: $1 0 0 2 \leq 1 1 1 2 \checkmark$
 $work = 1 1 1 2 + 1 3 5 9$
 $work = 2 14 6 6$
 P3: $0 0 2 0 \leq 2 14 6 6 \checkmark$
 $work = 2 14 6 6 + 0 6 3 2$
 $work = 2 10 9 8$
 P4: $0 6 4 2 \leq 2 10 9 8 \checkmark$
 $work = 2 10 9 8 + 0 0 1 4$
 $work = 2 10 10 12$
 P1: $0 3 3 0 \leq 2 10 10 12 \checkmark$
 $work = 2 10 10 12 + 1 4 2 0$
 $work = 3 14 12 12$

Yes, it can be fulfilled immediately.

Q4

Process	Allocated	Max	Available
	A B C D	A B C D	A B C D
P0	1 0 2 1	1 1 2 1	0 0 x 1 1
P1	2 0 1 1	2 2 1 0	
P2	1 1 0 1	2 1 3 1	
P3	1 1 1 0	1 1 2 1	

Find smallest value of x for which this is a safe state, give safe sequence.
 let x = 0

Process	Allocated	Max	Need	Available
	A B C D	A B C D	A B C D	A B C D
P0	1 0 2 1	1 1 2 1	0 1 0 0	0 0 0 1 1
P1	2 0 1 1	2 2 1 0	0 2 1 0 0	
P2	1 1 0 1	2 1 3 1	1 0 3 0 0	
P3	1 1 1 0	1 1 2 1	0 0 1 1 1	

- P0: $0 1 0 0 \leq 0 0 0 1 1 \times$
 P1: $0 2 1 0 0 \leq 0 0 0 1 1 \times$
 P2: $1 0 3 0 0 \leq 0 0 0 1 1 \times$
 P3: $0 0 1 1 1 \leq 0 0 0 1 1 \times$
 Impossible with x = 0,
 let x = 1

Process	Allocated	Max	Need	Available
	A B C D	A B C D	A B C D	A B C D
P0	1 0 2 1	1 1 2 1	0 1 0 0	0 0 1 1 1
P1	2 0 1 1	2 2 1 0	0 2 1 0 0	1 1 2 2 0
P2	1 1 0 1	2 1 3 1	1 0 3 0 0	
P3	1 1 1 0	1 1 2 1	0 0 1 1 1	

- P0: $0 1 0 0 \leq 0 0 1 1 1 \times$
 P1: $0 2 1 0 0 \leq 0 0 1 1 1 \times$
 P2: $1 0 3 0 0 \leq 0 0 1 1 1 \times$
 P3: $0 0 1 1 1 \leq 0 0 1 1 1 \checkmark$
 Impossible with x = 0,
 let x = 1

Process	Allocated	Max	Need	Available
	A B C D	A B C D	A B C D	A B C D
P0	1 0 2 1	1 1 2 1	0 1 0 0	0 0 1 1 1
P1	2 0 1 1	2 2 1 0	0 2 1 0 0	1 1 2 2 0
P2	1 1 0 1	2 1 3 1	1 0 3 0 0	
P3	1 1 1 0	1 1 2 1	0 0 1 1 1	

- P0: $0 1 0 0 \leq 0 0 1 1 1 \times$
 P1: $0 2 1 0 0 \leq 0 0 1 1 1 \times$
 P2: $1 0 3 0 0 \leq 0 0 1 1 1 \times$
 P3: $0 0 1 1 1 \leq 0 0 1 1 1 \checkmark$
 work = 0 0 1 1 1 + 1 1 1 0
 $work = 1 1 2 2 0$

- P0: $0 1 0 0 \leq 1 1 2 2 0 \times$
 P1: $0 2 1 0 0 \leq 1 1 2 2 0 \times$
 P2: $1 0 3 0 0 \leq 1 1 2 2 0 \times$
 P3: $0 0 1 1 1 \leq 1 1 2 2 0 \checkmark$
 $work = 1 1 2 2 0 + 0 0 1 1 1$
 $work = 1 1 3 3 1$

- P0: $0 1 0 0 \leq 1 1 3 3 1 \checkmark$
 $work = 1 1 3 3 1 + 1 0 2 1 1$
 $work = 2 1 5 4 2$
 P1: $0 2 1 0 0 \leq 2 1 5 4 2 \times$
 P2: $1 0 3 0 0 \leq 2 1 5 4 2 \checkmark$
 $work = 2 1 5 4 2 + 1 1 0 1 0$
 $work = 3 2 6 5 2$

- P0: $0 2 1 0 0 \leq 3 2 6 5 2 \checkmark$
 $work = 3 2 6 5 2 + 2 0 1 1 0$
 $work = 5 2 6 5 2$

Smallest possible value of x is 2